

Jesús M. Rueda-Becerril, Ph.D.

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Professional Summary

Computational physicist who turns complex physical models into fast, tested software and turns large scientific datasets into calibrated models. Ph.D. in Physics and ten years building simulation codes and the data pipelines around them: author of Paramo, an open-source Fortran/OpenMP radiative-transfer library used by other research groups; led a 7-person upgrade of the PatchworkMHD black-hole simulation code on the Frontera supercomputer; mapped wetland probability across a 2,160 km² watershed with random-forest models. Nine peer-reviewed articles and Co-I on a funded NASA Fermi Guest Investigator grant.

Skills

ML & Data: Scientific ML, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, random forest, SVM, neural networks, gradient-based optimization, parameter estimation, uncertainty quantification, synthetic dataset generation

Scientific Computing & HPC: Numerical PDEs (Fokker–Planck, advection–diffusion), radiative transfer, relativistic MHD, Lagrangian particle dynamics, finite-difference/finite-volume methods, numerical linear algebra, OpenMP, MPI, OpenACC, SLURM, PBS, HDF5

Geospatial & Remote Sensing: Sentinel-1/2 multispectral imagery, DEM terrain analysis, raster/vector pipelines, QGIS, geospatial feature engineering

Languages: Python, C/C++, Fortran, Rust, R, SQL, Java, Julia, Shell

Tools: Git, Docker, Azure, Jenkins, OpenShift, Kafka, Splunk, Jira, ParaView, VisIt

Experience

Independent Contractor

Jan 2026 – Present

Mercor

Remote

Physics subject-matter expert evaluating frontier language models for a leading AI lab.

- Reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology, and wrote prompts that stumped frontier models.
- Diagnose failure modes in model reasoning and write structured feedback to improve scientific accuracy.
- Benchmark model performance and quality-check solutions to complex STEM problems.

Spatial Data Scientist

May 2025 – Jan 2026

TealWaters

Seattle, WA

Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.
- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Engineered features from Sentinel-1/2 multispectral imagery and terrain derivatives for ML land-cover and wetland models; built Python processing tools and QGIS visualizations.
- Presented results to managers and decision-makers, trained 2 colleagues on the WIP tool, and worked with software engineers to move prototype tools into production.

Independent Researcher & Open-Source Developer

Jan 2024 – May 2025

Self-directed

Seattle, WA

Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, a Rust/Python toolkit that models radiation from relativistic outflows by solving the Fokker–Planck equation coupled to radiative transfer; published in ApJ (2024) [9].
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library that simulates wind-driven spore dispersal over forested, hilly terrain, parallelized with RcppParallel and OpenMP.
- Invited talk on machine learning in environmental science at the University of Washington (2024).

Software Engineer

Apr 2022 – Jan 2024

Paychex

Remote

Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Built Java Kafka consumers that moved data between enterprise databases, deployed through Jenkins and OpenShift CI/CD pipelines.

- Created Splunk dashboards tracking the users shared between enterprise systems and Oracle EBS, with alerts that flagged production incidents.
- Wrote full test suites for 5+ Python services in the CI pipeline, keeping production defect rates low.
- Prepared, validated, and analyzed Oracle EBS data with SQL.

Postdoctoral Research Associate

Rochester Institute of Technology

Feb 2021 – Apr 2022

Rochester, NY

HPC simulations of supermassive black-hole binaries in an NSF-funded project.

- Led a 7-person team upgrading PatchworkMHD, a C code for magnetohydrodynamic simulations of supermassive black-hole binaries.
- Added black-hole spin to the simulations with under 5% runtime overhead.
- Designed the simulation experiments and ran and benchmarked the code on TACC's Frontera supercomputer, identifying performance optimization opportunities.
- Calibrated Paramo model parameters against Fermi-LAT gamma-ray spectra with gradient-based optimization, and supervised a graduate student applying gradient descent to the same problem.
- Co-authored 2 papers from a multi-institution study of binary neutron-star mergers [6, 7] and a third on blazar heating and cooling [8].

Postdoctoral Research Fellow

Purdue University

Oct 2018 – Nov 2020

West Lafayette, IN

Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Architected Paramo, an open-source Fortran 95 library that solves the Fokker-Planck equation coupled to radiative transfer; parallelized it with OpenMP for a 20× speedup over the serial code on 10 cores.
- Paramo has been adopted by other research groups, including in Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals; one was funded (\$68K) and supported the finding that blazar jets are launched with similar energy per baryon, independently of their power [5].
- Added numerical-integration and OpenMP features to Paramo and built Python tools to compute energy losses and spectra of high-energy particles in gamma-ray burst afterglows [4].
- Mentored 3 graduate students (1 M.S., 2 Ph.D.), co-authoring papers with each.
- Wrote Python scripts for statistical modeling of the COVID-19 outbreak in Mexico with a multidisciplinary group, and co-wrote Spanish-language infographics and blog posts against misinformation.

Postdoctoral Research Fellow

Universidad Michoacana de San Nicolás de Hidalgo

Jan 2018 – Sep 2018

Morelia, Mexico

- Built a Python pipeline that turned GRTrans black-hole simulation images into training data for an SVM classifier of black-hole spin (grayscale conversion, intensity normalization, spatial feature extraction).
- Released an open-source Python tool for computing radiative-transfer spectra and light curves in relativistic astrophysics.
- Organized an HDF5 workshop to train graduate students in high-volume data storage.

Graduate Research Assistant

Universitat de València

Oct 2011 – Jul 2017

Valencia, Spain

- Automated runs of the C-SPEV radiative-transfer code with Shell and Python pipelines: 10–60 simulations per project and tens of GB of HDF5 output, with data-quality checks for downstream analysis.
- Analyzed multi-wavelength observations from the NASA Fermi-LAT telescope and the Very Long Baseline Array (VLBA), including cross-matching sources across catalogs by sky coordinates and fitting log-parabola spectra.
- Fit C-SPEV simulation models to telescope data, identifying spectral and light-curve patterns that quantify the magnetization of plasma in blazars.
- Extended C-SPEV to compute discrete and continuous spectra from particle distributions of arbitrary shape, without affecting simulation runtime.
- Published 2 first-author papers in MNRAS [2, 3].

Education

Ph.D. in Physics

Universitat de València

Oct 2011 – Jul 2017

Valencia, Spain

Excellent *cum laude*

Advisors: Prof. Miguel A. Aloy and Dr. Petar Mimica

Thesis: Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows

M.Sc. in Physics

Universidad Michoacana de San Nicolás de Hidalgo

Advisor: Prof. José A. Cervera

Thesis: *Study of TOV stars with the SPH method*

Aug 2009 – Sep 2011

Morelia, Mexico

B.Sc. in Physics

Universidad Autónoma del Estado de México

Advisor: Prof. Francisco S. Guzmán

Thesis: *Numerical solution of null geodesics for the generation of gravitational lenses produced by spherically-symmetric and static spacetimes*

Aug 2004 – Dec 2008

Toluca, Mexico

Lic. Juan Josafat Pichardo Cruz Award

Publications

Articles (9 published, 3 first-author; 3 in preparation)

- [12] Winkowski, S., **Rueda-Becerril, J. M.**, & Willing, C. *Using Numerical Simulations of Wind on Hilly Terrain to Model Propagule Dispersal*, in prep.
- [11] Holzman, L., **Rueda-Becerril, J. M.**, Creamer, C., & Willing, C. *Random Forest Link Landscape Position to Drought Resilience in Giant Sequoia Forests*, in prep.
- [10] **Rueda-Becerril, J. M.**, Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.
- [9] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.
- [8] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS 513, 5766–5779 (2022), arXiv:2201.07790.
- [7] Lopez-Armengol, F. G., Etienne, Z. B., [...], **Rueda-Becerril, J. M.**, [...] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, Phys. Rev. D 106, 083015 (2022), arXiv:2112.09817.
- [6] Murguia-Berthier, A., Noble, S., [...], **Rueda-Becerril, J. M.**, [...] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, ApJ 919, 95 (2021), arXiv:2106.05356.
- [5] **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.
- [4] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.**, & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, MNRAS 496, 974–986 (2020), arXiv:1910.14049.
- [3] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.
- [2] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, MNRAS 438, 1856–1869 (2014), arXiv:1310.5441.
- [1] Guzmán, F. S., & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, Phys. Rev. D 80, 084023 (2009), arXiv:1009.1250.

Conference Proceedings

- [P5] **Rueda-Becerril, J. M.** *A numerical approach for radiative cooling in relativistic outflows*, Astron. Nachr. 342, 277–282 (2021), 9th International Workshop on Astronomy and Relativistic Astrophysics, arXiv:2011.13797.
- [P4] **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *The blazar sequence revised*, Astron. Nachr. 342, 147–152 (2021), 9th International Workshop on Astronomy and Relativistic Astrophysics, arXiv:2011.13805.
- [P3] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *Numerical simulations of the internal shock model in magnetized relativistic jets of blazars*, PoS(SWIFT 10) 233, 159 (2014), arXiv:1502.07882.
- [P2] **Rueda-Becerril, J. M.**, Mimica, P., Aloy, M. A., & Aloy, C. *Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets of blazars*, EPJ Web Conf. 61, 02007 (2013), arXiv:1309.4612.
- [P1] Mimica, P., Aloy, M. A., **Rueda-Becerril, J. M.**, Tabik, S., & Aloy, C. *Numerical simulations of dynamics and emission from relativistic astrophysical jets*, J. Phys.: Conf. Ser. 454, 012001 (2013), arXiv:1211.1794.

Grants and Fellowships

Research Grants

NASA Fermi Guest Investigator Program, Cycle 12

A simple model to understand the blazar sequence

PI: D. Giannios; Co-I: J. M. Rueda-Becerril

2019

Grant #121077, \$68K

Fellowships

International Postdoctoral Fellowship <i>National Council of Science and Technology (CONACyT), Mexico</i>	Oct 2018 – Nov 2020
Postdoctoral Fellowship <i>Secretariat of Public Education, Mexico (Program for the Professional Development of Higher Education Institutions)</i>	Jan 2018 – Sep 2018
Study-Abroad Fellowship <i>National Council of Science and Technology (CONACyT), Mexico</i>	Sep 2014 – Aug 2016
Santiago Grisolia Fellowship <i>Council of Education, Research, Culture and Sport, Valencian Community, Spain</i>	Oct 2011 – Jun 2014
M.Sc. Fellowship <i>CONACyT, at the Institute of Physics and Mathematics, Universidad Michoacana de San Nicolás de Hidalgo</i>	Sep 2009 – Aug 2011
Summer Research Fellowship (XVII Summer of Scientific Research) <i>Mexican Academy of Sciences</i>	Jun 2007 – Aug 2007

Software and Projects

Blazar Population ML <i>Scientific ML (Python, scikit-learn, PyTorch)</i>	Jan 2026 – Present github.com/altjerue/ilwikak-ml
- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data. - Building a neural network that maps physical jet parameters to observable spectral energy distributions accurately enough to compare with Tleco results. - Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves.	
WindsOfChange <i>Lagrangian particle-dispersion library (C++/R)</i>	Aug 2024 – Present github.com/UW-ForestMycobiomeLab/WindSporesHills
Simulates particle dispersion, atmospheric turbulence, and flow over hilly terrain with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004); designed to scale to tens of thousands of particles over kilometer-scale domains.	
Tleco <i>Radiative-transfer toolkit (Rust/Python)</i>	Jan 2024 – Sep 2024 github.com/zkdavis/Tleco
Numerical PDE solvers, particle-distribution algorithms, and spectral calculations for relativistic plasmas; successor to Paramo [9].	
Paramo <i>Radiative-transfer library (Fortran 95, OpenMP)</i>	Oct 2018 – Apr 2022 github.com/altjerue/paramo
Fokker–Planck and radiative-transfer solver for relativistic particle distributions; 20× OpenMP speedup on 10 cores; used by other groups, including Combi & Siegel (2023).	
PatchworkMHD <i>Large-scale MHD simulation code (NSF project)</i>	Feb 2021 – Apr 2022
Led a 7-person upgrade of this production C code for HPC environments, adding black-hole spin with under 5% runtime overhead.	

Awards

Marcos Moshinsky Award for Best Poster <i>9th International Workshop on Astronomy and Relativistic Astrophysics (IWARA 2020), video conference</i>	Sep 2020
Lic. Juan Josafat Pichardo Cruz Award <i>Universidad Autónoma del Estado de México</i> For defending a licentiate thesis within a year of completing the undergraduate coursework.	2009

Presentations

Invited Talks

The Role of Machine Learning in Environmental Science <i>Seminar on Climate Change Ecology, University of Washington</i>	Mar 2024 Seattle, WA
Morphology of the spectra from numerical simulations of the internal shocks model for blazars <i>Astrophysics Seminar, Purdue University</i>	Feb 2019 West Lafayette, IN

Numerical simulations of the internal shocks model in magnetized relativistic jets of blazars Jun 2018
DATA Group Weekly Seminar, Instituto de Astronomía, UNAM Mexico City, Mexico

Numerical treatment of non-thermal radiation in the internal shocks model for blazars Mar 2018
Weekly Seminar, Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo Morelia, Mexico

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars Oct 2014
IVICFA's Fridays: Computation in Physics, IFIC Paterna, Spain

Contributed Talks

Simulations of supermassive binary black holes accretion dynamics in the spinning case Apr 2022
APS April Meeting 2022 New York, NY

A numerical approach to the Klein-Nishina corrections of radiative cooling in relativistic outflows Apr 2021
APS April Meeting 2021 Online

The blazar sequence revised Sep 2020
9th International Workshop on Astronomy and Relativistic Astrophysics Online
 Video: youtube.com/watch?v=BAZNWLNT69M

Influence of the magnetic field on the spectral properties of blazars in the internal shocks scenario Jun 2014
Extreme-Astrophysics in an Ever-Changing Universe Ierápetra, Greece

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets Jul 2013
XXXIV Biennial Meeting of the Royal Spanish Society of Physics Valencia, Spain

Posters

A numerical approach for radiative cooling in relativistic outflows Sep 2020
9th International Workshop on Astronomy and Relativistic Astrophysics Online
 Marcos Moshinsky Award for Best Poster. Video: youtube.com/watch?v=0TJiKg7kOPi

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars Dec 2014
Swift: 10 Years of Discovery Rome, Italy

Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets Jun 2013
The Innermost Regions of Relativistic Jets and Their Magnetic Fields Granada, Spain

Teaching and Mentoring

Zachary Davis, graduate student 2018 – 2022
Department of Physics and Astronomy, Purdue University Mentoring
 Co-authored papers [8, 9]

Amanda O. Harrison, graduate student 2018 – 2020
Department of Physics and Astronomy, Purdue University Mentoring
 Co-authored paper [5]

Hao Zhang, graduate student 2018 – 2019
Department of Physics and Astronomy, Purdue University Mentoring
 Co-authored paper [4]

Thermodynamics (graduate level) Jun 2018
Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo Substitute Instructor

Professional Development

High Performance Computing on Frontera May – Jun 2021
Jason Allison et al., TACC, Austin, TX Lecture series

Writing Winning Grants Nov 2019
Dr. Lauren Broyles, Purdue University Lecture

XSEDE HPC Workshop: Summer Boot Camp Jun 2019
John Urbanic, Purdue University Workshop

Data Analysis and Machine Learning with Python Feb 2017
Dr. Alejandro Torres, Universitat de València Workshop

Numerical Relativity Simulations of BBH Coalescence Using the Einstein Toolkit Jul 2016
Dr. Vassilios Mewes, Universitat de València Workshop, 8 hours

The Universe in the Light of PLANCK and BICEP2 <i>Prof. Nick Mavromatos, Universitat de València</i>	May 2014 Lecture series, 2 credits
Dark Matter <i>Prof. Alejandro Ibarra, Universitat de València</i>	Sep 2013 Lecture series, 2 credits
International Cargèse School on Cosmic Accelerators <i>Institut d'Études Scientifiques de Cargèse, France</i>	Apr – May 2013 Summer school
Introduction to C++ Programming <i>Dr. Jacek Generowicz, Universitat de València</i>	Apr 2012 Workshop, 6 credits
Numerical Relativistic Astrophysics <i>Prof. Luciano Rezzolla, Universitat de València</i>	Mar – Apr 2012 Lecture series, 9 hours
Fortran for Scientific Computing <i>HLRS, University of Stuttgart, Germany</i>	Mar 2012 Workshop, 33 hours

Certifications

Gen AI Intensive <i>Generative AI</i>	Kaggle
Data Scientist Professional with Python <i>Python programming, data science, data communication, machine learning</i>	DataCamp
Machine Learning Scientist with Python <i>Machine learning, NLP, deep learning, image processing, big data</i>	DataCamp
Mathematical Foundations of Machine Learning <i>Python programming, data science, statistics, machine learning</i>	Udemy
Python for Statistical Analysis <i>Python programming, data science, data communication, machine learning</i>	Udemy

Outreach

Los más rápidos y los más furiosos (The Fastest and the Most Furious) <i>Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco</i>	Sep 2020 Online talk
Una simulación de la física y la astrofísica (A Simulation of Physics and Astrophysics) <i>Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco</i>	Aug 2020 Online talk
ANITA y la teoría de los universos paralelos (ANITA and the Theory of Parallel Universes) <i>Científicos Mexicanos en el Extranjero</i> mexiciencia.github.io/post/anita	May 2020 Blog post
¿Qué es el modelo SIR? (What is the SIR Model?) <i>Científicos Mexicanos en el Extranjero</i> mexiciencia.github.io/post/modelo-sir	May 2020 Blog post
Evolución del brote epidémico de COVID-19 (Evolution of the COVID-19 Epidemic Outbreak) <i>Científicos Mexicanos en el Extranjero</i> Data analysis and modeling collaborator. mexiciencia.github.io/post/covid19	Apr 2020 Blog post
Annual Department of Physics and Astronomy Poster Event <i>Purdue University</i>	Nov 2019 3 posters
Post-Doc Panel Q&A: What Happens When We Complete Our PhDs? <i>Department of Physics and Astronomy, Purdue University</i>	Apr 2019 Panelist
Annual Department of Physics and Astronomy Poster Event <i>Purdue University</i>	Nov 2018 Poster
¿Decía Einstein la verdad? (Was Einstein Telling the Truth?) <i>Facultad de Ciencias, Universidad Autónoma del Estado de México</i>	Mar 2009 Talk

Service

Organizing contributor <i>X Scientific Meeting of the Spanish Astronomical Society</i>	Dec 2012 Valencia, Spain
Student representative, Governing Council of the Faculty of Sciences <i>Universidad Autónoma del Estado de México</i>	Aug 2007 – May 2009 Toluca, Mexico

Languages

Spanish (native), English (full professional proficiency), Catalan (basic), Nahuatl (basic)